

Networking Concepts in the Cloud

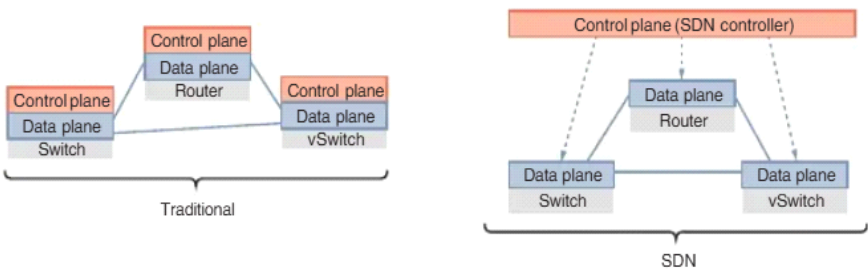
Cloud networking works **like traditional data center networking**, but it happens mostly on a **logical (virtual)** level, not physical hardware.

Cloud networking = same logic as physical networks, but everything is virtual and managed through cloud platforms.

Networking Concepts

- **SDN (Software-Defined Networking)**: separates how the network is controlled (control plane) from how data moves (data plane).
- **NFV (Network Functions Virtualization)**: virtualizes network devices and services, taking SDN further.
- The same networking technologies are used, but they are now **virtual** instead of physical.

In short: Cloud networking replaces physical hardware with software-based, virtualized systems for flexibility and control.



Traditional = each device controls itself.
SDN = one central brain controls everything.

Figure 4-1 Individual control planes in a traditional network versus a centralized control plane in an SDN network

- In a **traditional network**, each device (router, switch, etc.) has its **own control plane** — it manages itself.
- In **SDN (Software-Defined Networking)**, the **control plane is centralized** in one controller that manages all devices.

From OSI Model to Cloud Stack

1. **Cloud Stack:** A 5-layer model for cloud, defined by who has access:
 1. **Physical Layer:** The physical infrastructure (CSP's responsibility).
 2. **Virtualization Layer:** The hypervisor layer (CSP's responsibility).
 3. **Network Layer:** The IaaS layer where customers create and manage virtual networks (e.g., VPCs, subnets).
 4. **OS Layer:** The PaaS layer where customers manage OS environments.
 5. **Application Layer:** The SaaS layer where users access applications.



TCP/IP model		OSI model
4. Application layer	Interface with applications	7. Application layer
	Data formatting	6. Presentation layer
	Conversation and session management	5. Session layer
3. Transport layer	Delivery and sequencing of data (Port number)	4. Transport layer
2. Network layer	Logical addressing (IP address)	3. Network layer
1. Link layer	Physical addressing (MAC address)	2. Data link layer
	Physical connection	1. Physical layer

Figure 4-3 The OSI and TCP/IP models of networking communications

Software-Defined Networking in the Cloud

- **SDN (Software-Defined Networking)** is the **foundation of cloud networking**.
- Users **can't access physical hardware**, but they **control the network through software**.
- In **public clouds**, the **cloud provider** manages the physical infrastructure (fiziksel altyapı). and ensures availability.
- **High availability** is achieved by using **redundant (duplicate) virtual devices**.

In short: SDN lets users manage cloud networks with software — not hardware — while the cloud provider handles the physical part.

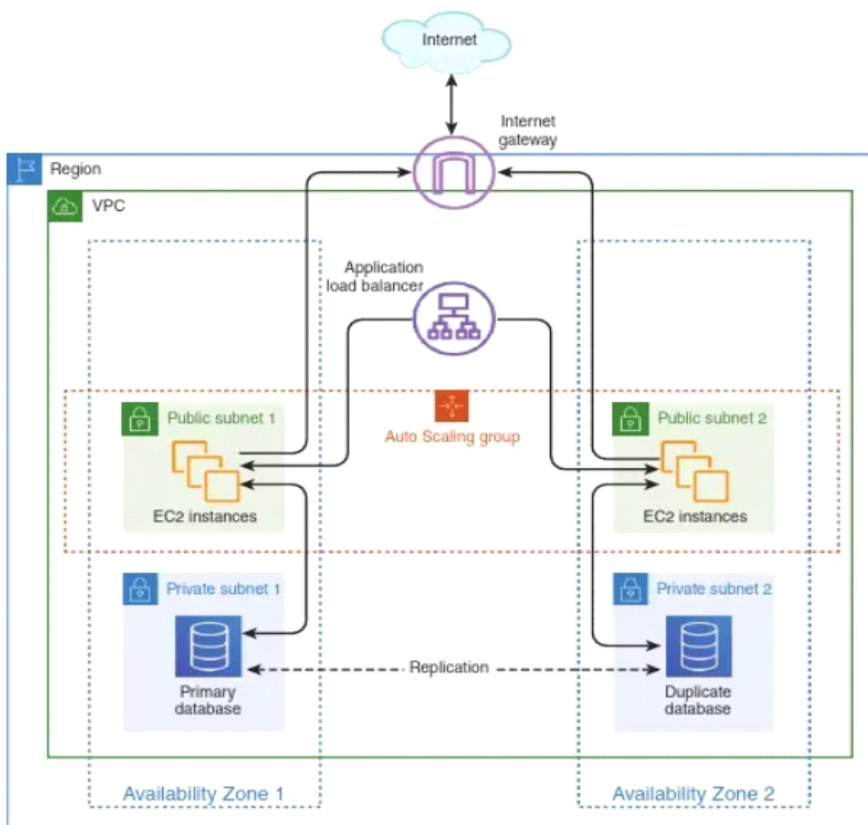


Figure 4-5 Redundant EC2 instances and databases

- Cloud systems use **two availability zones** to keep services online.
- **Load balancer** sends traffic to healthy EC2 instances.
- **Auto scaling** adds or removes servers when needed.
- **Databases** are copied between zones to stay available if one zone fails.

This setup keeps the system running without crashes or downtime.

If one data center (for example, Zone 1) goes down, **Zone 2** takes over.

The users do not notice anything, and the system keeps working normally.

! IP Address Space

- Every device in a network has an **IP address** to identify it.
- Cloud resources have **two types** of IPs: **public** and **private**.
- A **public IP** lets you connect from the Internet.
- A **private IP** is used only inside the cloud network.

IP Addressing

- In local networks, a **DHCP server** gives private IP addresses to devices.
- In the cloud, IPs are given **automatically** from a set of defined ranges.
- The **IETF**(Internet Engineering Task Force) created **RFC(Request For Comments) 1918** to show the best way to use private IP addresses.

! Subnetting

Subnetting (1 of 2)

- **CIDR notation** is used to write IPv4 address blocks, so subnet masks are not needed.
 - Example: 192.168.0.0/16
 - The number after the slash shows how many bits are for the **network ID**.
- Example: 192.168.4.1 in binary is 11000000.10101000.00000100.00000001.
 - Each part has **8 bits** (called an **octet**).
 - If the address is /24, the first **24 bits** show the network, and the last **8 bits** show the **host**.
- If more bits are used for the **network ID**, fewer bits are left for **hosts**.
- Dividing an IP range like this is called **subnetting**.
- There are **three main subnet classes**:
 - **Class A**: First octet → 10.0.0.0/8
 - **Class B**: First two octets → 172.16.0.0/16
 - **Class C**: First three octets → 192.168.100.0/24
- Some IPs in each subnet are **reserved** and **cannot be used** for hosts.
- Class A → /8 → 8 network bits, 24 host bits
- Class B → /16 → 16 network bits, 16 host bits
- Class C → /24 → 24 network bits, 8 host bits

Cloud Network Interfaces

- Every computer or device interface needs its own **IP address** and can have different settings.
- When you make a **Virtual Machine (VM)**, it also gets a **virtual network interface (vNIC)** with its own ID and configuration.
- The **vNIC** connects the VM to its **subnet** inside the virtual network.
- To make the network faster, a method called **SR-IOV (Single Root Input/Output Virtualization)** is used.
 - This lets the **host** send network traffic **directly to the NIC hardware**, instead of processing it through a **virtual switch created by hypervisor**.

! Networking in AWS

- When you make a network in the cloud, you create your own private area called a **VPC (Virtual Private Cloud)**.
 - In Microsoft Azure, this is called a **VNet**.
- When you open a new AWS account, AWS automatically makes a **default VPC** for you.
- When you create a **VM** in AWS, it is automatically set to use this default VPC (you can change it if needed).
- Inside the VPC, you can set up **subnets, route tables, gateways, and network connections**.

! Regions in AWS

- **AWS** has data centers in **more than 24 regions** around the world.
- Each **region** is named after its **geographical location** and has a **unique code** (for example, us-east-1).
- When you make an **AWS account**, you get a **default region**, but you can **change it**.
- You might change your region for these reasons:
 - **Cost:** Prices are different in each region.
 - **Performance:** Some regions are **faster** or have **lower latency**.
 - **Features:** Not every AWS service is available in all regions.

- Each **AWS region** has several **Availability Zones (AZs)**.
- An **Availability Zone** is a separate physical data center inside a region.
- Using **multiple AZs** gives **high availability** — if one zone fails, another keeps running.
- This helps provide **redundancy** and **reliable cloud performance**.



Figure 4-7 AWS public regions in North America

Load Balancer mission :

- **Distributes incoming traffic** evenly across multiple servers (EC2 instances).
→ So no single server gets overloaded.
- **Provides High Availability (HA)** by automatically sending traffic only to **healthy** servers.
→ If one server (or subnet) goes down, the load balancer reroutes traffic to others.

VPCs in AWS

- Each **default VPC** gets a **/16 IPv4 CIDR range** automatically. You can make **smaller custom VPCs** if you want.
- Every **VPC** includes **all Availability Zones (AZs)** in its region.
- **DNS** and **DHCP** settings are first made at the **VPC level**.
- When creating a new VPC, you can choose the **CIDR range** using the **VPC IP Address Manager (IPAM)** tool.
- **IPAM (IP Address Management)** helps manage all IP addresses more easily and efficiently across many VPCs.

Public subnets connect to the Internet Gateway.

Private subnets do not — they're isolated and used for backend resources.

- Instances here (like EC2 VMs) can directly talk to the Internet.
- Usually used for things like:
 - Web servers
 - Load balancers
 - Bastion hosts (for SSH access)

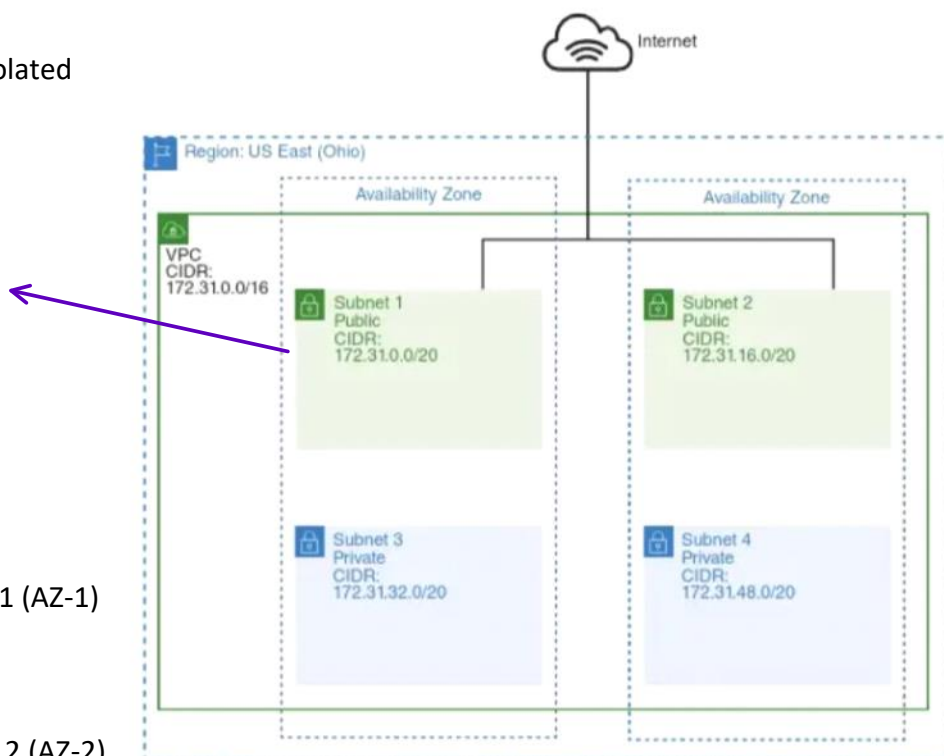


Figure 4-12 Public subnets in AWS have access to the Internet, whereas private subnets do not

Two AZs × (Public + Private) = **4 subnets for redundancy and security.**

- **Subnet 1 (Public, AZ-1)** → EC2 instance running your app.
- **Subnet 2 (Public, AZ-2)** → another EC2 instance, *exact same app*.

Subnet	AZ	Type	CIDR	Internet Access	Typical Use
Subnet 1	AZ-1	Public	172.31.0.0/20	✔ Yes	Web Server
Subnet 2	AZ-2	Public	172.31.16.0/20	✔ Yes	Load Balancer
Subnet 3	AZ-1	Private	172.31.32.0/20	✗ No	Database
Subnet 4	AZ-2	Private	172.31.48.0/20	✗ No	Backup DB / Internal App

! Gateways and Route Tables in AWS

Internet Gateway (IGW or IG)

- It's the **door from your VPC to the Internet**.
- Used by **public subnets** so their EC2 instances can send and receive data online.
- It provides NAT (Network Address Translation) : **hide private IP addresses** behind one **public IP** when sending traffic to the Internet.
- The route table usually has:
0.0.0.0/0 → Internet Gateway
→ means "send all Internet traffic through the gateway."

NAT Gateway (NGW)

- Used by **private subnets**.
- Allows instances *inside* the private subnet to **go out** to the Internet (e.g., for software updates).
- But it **blocks incoming connections** — the Internet can't reach them directly.
- So it's for **outbound only** traffic (egress).

Route Table

- Route Table controls where traffic goes (inside VPC and comes through to VPC).
- Example routes:
 - 172.31.0.0/16 → local (traffic stays inside VPC)
 - 0.0.0.0/0 → igw-xxxx (Internet traffic goes to IGW)

Internet Gateway (IGW)

- Shared by all subnets.
- Public subnets use it directly, private subnets use it indirectly through NAT.
- Without IGW → no Internet access, period.

Let's say one EC2 in a private subnet needs to download an update:
EC2 (private) → NAT Gateway (public subnet) → Internet Gateway → Internet

for incoming requests to a web app:
Internet → Internet Gateway → Public Subnet (EC2 or Load Balancer)

Public Subnets (Subnet 1 & 2)

- Each AZ has a **public subnet**.
- Inside each public subnet sits a **NAT Gateway**.
- They're connected to the **Internet Gateway (IGW)** — the "door" to the outside world.

Private Subnets (Subnet 3 & 4)

- Each AZ also has a **private subnet** (no direct Internet connection).
- They contain **EC2 instances** (your actual servers).
- These EC2s connect to the **NAT Gateway** in their AZ to reach the Internet (for updates, API calls, etc.).
- Route table example for Subnet 3 & 4:

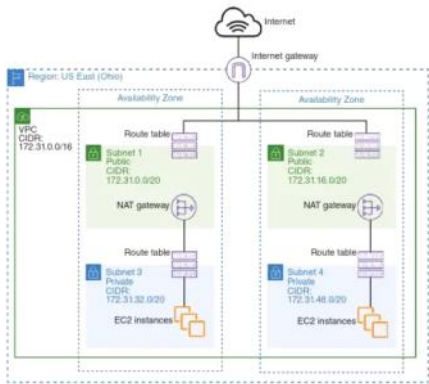


Figure 4-13 An Internet gateway, NAT gateways, and route tables in AWS

Private subnets can go *out* to the Internet via **NAT**, but no one can reach them *from* the Internet.

This diagram shows **how public and private subnets communicate** with the Internet using **Internet Gateways, NAT Gateways, and Route Tables** inside a VPC.

each AZ (left & right) is isolated = high availability

Subnet 1 & Subnet 2 (Public Subnets)

Main mission:

- Let Internet traffic in/out — they act as gateways for communication between your private network and the outside world.

They host:

- NAT Gateways → so private subnets can go online safely
- Load Balancers or Public EC2s (like web servers) that need to be reachable from the Internet

They connect directly to:

- Internet Gateway (IGW)

In short:

Subnet 1 and 2 are like the "front doors" of your cloud network — handling Internet access and protecting private subnets behind them.

NAT Gateways (NGW)

- Private instances sometimes need to **access the Internet** (e.g., download updates).
- They do this by sending their traffic → **to a NAT Gateway** → which sits in a **public subnet** → and sends it out through the **Internet Gateway**.
- Private instances **stay safe** — no one from the Internet can reach them directly.
- They can only **initiate** traffic **outward** (never get traffic in).
- You need **fewer public IPs** because many private instances share one NAT Gateway.
- Easier to **monitor & filter** what traffic leaves your private network

Three-Tier Architecture in the Cloud !!

It's a way of organizing applications into **three layers (tiers)** for better performance, security, and scalability.

1. Presentation Tier

- This is the **front end** — what users interact with.
- Example: browsers, mobile apps, or web interfaces.

2. Logic Tier

- The **backend application** — where business logic runs.
- It handles user requests, processes data, and talks to the database.
- Example: a web server or API hosted on an EC2 instance.

3. Data Tier

- This is where **databases** live.
- It's **not directly accessible from the Internet** — only the logic tier can communicate with it.
- Example: an RDS (database) instance.

In short:

Frontend (presentation) → Backend logic (application) → Database (data)
Each tier has its own subnet and security rules.

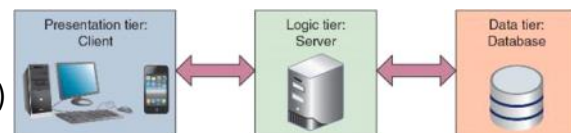


Figure 4-15 Basic three-tier architecture

Regions and Availability Zones in Azure

• Azure Regions:

Microsoft has about **60 regions worldwide** — each region is a separate geographic area (like “East US”, “West Europe”) that contains one or more data centers.

• Geographies:

Regions are grouped into **geographies**, which are bigger areas (like “North America” or “Europe”) that ensure **data compliance** and **residency** rules are followed.

• Availability Zones (AZs):

Not all regions have AZs.

Regions that do have **at least three AZs**, and each AZ is made up of one or more **physically separate data centers** with independent power and networking.

• High Availability:

If you deploy across at least **two AZs**, Microsoft guarantees **99.99% uptime** — meaning even if one data center fails, your services stay online.

! Route Tables in Azure

- Azure creates system routes automatically when you make a Virtual Network (VNet).
- These system routes allow communication between all resources in the VNet.
- You cannot change system routes.
- You can make your own route table to override the default routes.
- Azure does not use Internet Gateways (IGs) like AWS.
- You can create a NAT gateway using the Azure Virtual Network NAT service for outbound Internet access.

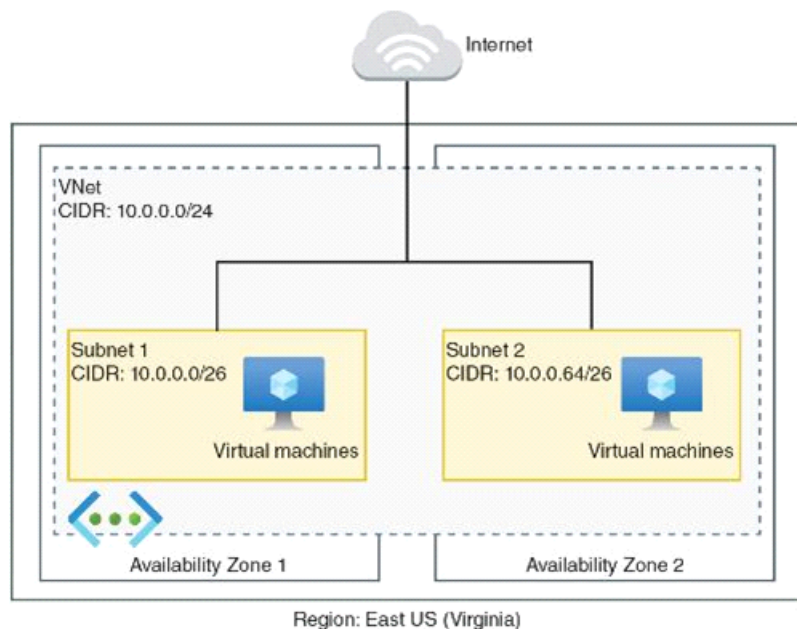


Figure 4-19 Subnets in an Azure VNet can communicate with one another by default

! Networking in GCP

- GCP works in a similar way to AWS.
- There are some key differences:
 - A GCP VPC is **global** and covers all regions by default.
 - A GCP VPC does not **have its own IP range**; CIDR ranges are set at the subnet level. (bölgeye özel IP range subnet'ler ile sağlanmıştır.)
 - IPv6 traffic works only for external resources, not between internal GCP VPC resources.

- VPC = global “kapsayıcı” ağ (IP yok)
- Subnet = bölgesel ağ (IP aralığı burada tanımlanır)

VM → Internet ✓
 Internet → VM ✓
 VM → VM ✗ (IPv4)

! Regions and Zones in GCP

- GCP has about 30 regions around the world.
- Most regions include at least three zones with separate power, cooling, networking, and control systems.
- A VM in one region can talk to a VM in another region without using the public Internet.
 - This improves security.
- When choosing regions and zones, keep your resources close to the users who will access them often.

! VPCs and Subnets in GCP

- A GCP VPC covers (spans) more than one region.
- If you create a VPC in **auto mode**, Google Cloud makes one **subnet in every region** automatically.
- Auto mode networks use **predefined IP address ranges**.
 - You can also choose **custom mode** to decide where your subnets will be.
- A global VPC lets you **reach far regions** from your on-premises (local) network through **one single VPC connection**.

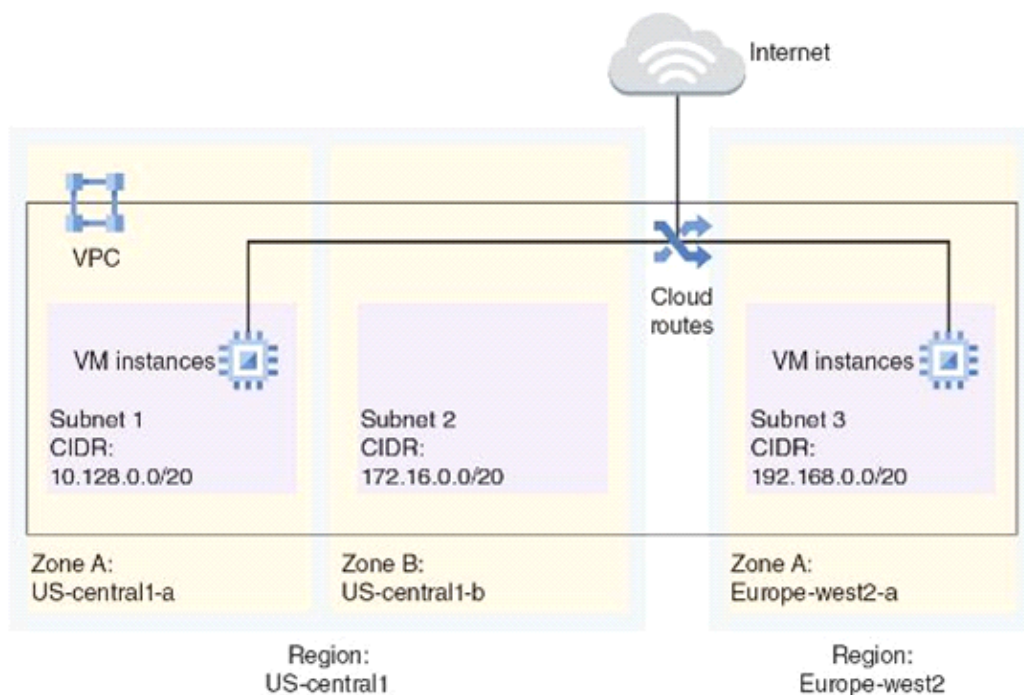


Figure 4-22 A single VPC can span all of GCP's regions

One single GCP VPC can connect **different regions and zones** around the world.

- **Region US-central1** (in the U.S.) has:
 - **Subnet 1** → CIDR: 10.128.0.0/20 → Zone A (US-central1-a)
 - **Subnet 2** → CIDR: 172.16.0.0/20 → Zone B (US-central1-b)
- **Region Europe-west2** (in Europe) has:
 - **Subnet 3** → CIDR: 192.168.0.0/20 → Zone A (Europe-west2-a)

How they connect

- All three subnets (U.S. and Europe) are part of **one VPC**.
- They are connected through **Google's private global network**, not the public Internet.
- **Cloud routes** inside the VPC allow VM instances in different regions to communicate safely and quickly.

Internet access

If any VM needs to reach the Internet, it does so through the **VPC's Internet gateway** — but internal traffic between subnets **stays private** inside Google's backbone.

Main idea

A **single global VPC** in GCP can include **subnets across many regions**, and all of them can talk to each other directly without using the public Internet.

Routes in GCP

- Routes in GCP work at the **VPC level**, not at the subnet level.
- Each VM inside a subnet keeps its **own copy** of the route table.
- When you create a route table, GCP makes two main types of routes:
- **1. System-generated routes**
 - **Subnet routes:** let traffic move between all subnets in the same VPC.
 - **Default route:** sends all other (external) traffic to the Internet gateway.
- **2. Custom routes**
 - **Dynamic routes:** made automatically by the *Cloud Router* service.
 - **Static routes:** made manually by the user.